Reza Sajjadinasab

PhD Candidate, Computer Engineering | Machine Learning, Reinforcement Learning & Compilers

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SUMMARY

Computer engineering PhD candidate at Boston University working across machine learning and systems. I fine-tune pretrained vision policies for autonomous driving with reinforcement learning (92.03 CARLA Bench2Drive driving score, comparable to the strongest published camera-only policy), and with **Red Hat** mentors I build learned models that optimize compilers (first pick beats GCC `-Oz` before anything is built). Comfortable from PyTorch research code down to CUDA, AVX, and FPGA. 8 papers (4 published, 4 under review). Open to internships and full-time roles in ML, RL, computer vision, compilers, and high-performance computing. Graduating August 2027.

TECHNICAL SKILLS

Machine Learning PyTorch, TensorFlow, scikit-learn, LightGBM; reinforcement learning (SAC, PPO, constrained RL), imitation learning, world models, vision transformers (DINOv3), transformers, ensembles and uncertainty, Bayesian optimization

Compilers & Performance GCC flag tuning and autotuning, learned cost models, CUDA kernels, AVX/SIMD intrinsics, OpenMP, roofline analysis, benchmarking

Hardware & Systems Verilog, FPGA, HLS (Xilinx Vitis, Intel HLS), Gem5, Intel Pin; asynchronous multi-GPU training, sharded and resumable data pipelines; LLM agents with retrieval (FAISS)

Languages & Tools Python, C/C++, CUDA, Java, SQL, MATLAB, Verilog, Linux, Git

RESEARCH & ENGINEERING EXPERIENCE

Graduate Researcher — Reinforcement Learning for Autonomous Driving **Boston University**
Advisor: Prof. Eshed Ohn-Bar | PyTorch, CARLA, HUGSIM Boston, MA | 2024 – Present

- Designed and built **Greg**, off-policy RL fine-tuning of a 361M-parameter camera-only driving policy (frozen DINOv3 encoder): constrained SAC anchored to the imitation policy, distributional critic ensembles, and world-model value expansion (under review).
- Raised Bench2Drive driving score from **84.28 to 92.03**, comparable to the strongest published camera-only policy; **zero collisions** on 28 adversarial routes (54.26 vs 27.96 for the best imitation baseline) and zero-shot transfer to HUGSIM.
- Built the asynchronous training system on 2× RTX 3090 (simulators feeding a shared replay buffer, learner on the second GPU; 480 GPU hours per run); the policy runs at 27.4 FPS on one GPU.

Graduate Researcher — ML-Driven Compiler Optimization **Boston University | Red Hat**
Advisor: Prof. Martin Herbordt | Mentors: Sanjay Arora and Ulrich Drepper, Red Hat Boston, MA | 2022 – Present

- Built **Flagwise**, a transformer cost model over code-LM embeddings that picks GCC flags: its **first pick, made before anything is compiled, is 1.03% smaller than `-Oz`** on 91 C programs, and 20 builds reach **6.25%**, worth 169 random builds and above SMAC3, BOCS, and OpenTuner (under review).
- Built **FlagPool**, **182,000 correctness-gated builds** from a sharded, resumable driver (17.4 builds/s on 28 workers); to be released.
- Earlier: 8–13% smaller code than `-Oz` with 15 fewer compilations than OpenTuner (IEEE HPEC 2024); hand-vectorized Smith–Waterman with AVX for **6**× over naive (IEEE IPDPSW 2024).

Ansible Wrangler — LLM Incident-Remediation Agent (course project) **Boston University, EC528 Cloud Computing**
RAG, self-hosted LLM, Ansible/AWX | Industry mentor: Ryan Cook, Red Hat Boston, MA | Fall 2024

- Course project, 6-person team: an LLM agent that turns ServiceNow incidents into deployed Ansible playbooks via FAISS retrieval, a self-hosted Qwen2.5-Coder-32B, automated GitHub pull requests, and AWX validation.

Quantitative Developer & Software Contractor (Freelance) **Independent**
Statistical modeling, algorithmic trading, CUDA Remote | 2020 – 2021

- Built and deployed forecasting-based trading strategies in MQL5; ported a Monte Carlo ray tracer to CUDA, replacing a multithreaded CPU renderer in a production pipeline.

EDUCATION

Ph.D., Computer Engineering, Boston University — Co-advisors: Profs. Herbordt & Ohn-Bar Boston, MA | 2022 – Aug 2027 (expected)

M.S., Computer Engineering, Boston University — Advisor: Prof. Martin Herbordt Boston, MA | 2022 – 2024

B.S., Electrical Engineering, Sharif University of Technology — Advisor: Prof. Iman Gholampour Tehran, Iran | 2017 – 2022

PUBLICATIONS & HONORS

- **8 papers** (4 published, 4 under review), including IEEE HPEC '24, IEEE IPDPSW '24, *IET ITS* '22, and *ASCE JTE* '24. Google Scholar
- Distinguished Computer Engineering Fellowship, BU, 2022; top 0.1% (192nd of ∼200,000) in Iran's national entrance exam, 2017